Literature resolution of the Holevo–Utkin sharp zero-sum norm conjecture

Status note for run `fa_banach_001`

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Status: literature already answered. Haonan Zhang explicitly identifies the Holevo–Utkin conjecture and proves every dimension not already proved in the source paper.

Original conjecture

The source is A. S. Holevo and A. V. Utkin, *A conjecture on a tight norm inequality in the finite-dimensional ℓ_p*, arXiv:2603.24017 [1]. On PDF page 2, equations (1)–(4) state the following. Put

$$L = \left\{ x \in \mathbb{R}^d : \sum_{i=1}^{d} x_i = 0 \right\}, \qquad d \geq 3.$$

For $\alpha \geq 1$ they conjecture

$$\|x\|_{2\alpha} \leq M(d,\alpha)^{1/(2\alpha)} \|x\|_2, \qquad x \in L,$$

while for $0 < \alpha < 1$ the inequality is reversed. The exact proposed constant is

$$M(d,\alpha) = \max\bigl\{2^{1-\alpha}, d^{-\alpha}\bigl((d-1)^{\alpha} + (d-1)^{1-\alpha}\bigr)\bigr\}$$

for $\alpha > 1$, and the corresponding minimum for $1/2 < \alpha < 1$; for $0 < \alpha \leq 1/2$ it is $2^{1-\alpha}$. The source writes the same choice using its transition dimension $d(\alpha)$. The two displayed values come from the normalized forms of

$$(1, -1, 0, \ldots, 0), \qquad (d-1, -1, \ldots, -1).$$

The source proves $d = 3$ and reports numerical verification through $d = 200$.

Exact later answer

Haonan Zhang, *Proof of the Holevo–Utkin conjecture on sharp ℓ_p norms for zero-sum vectors*, arXiv:2605.05243v2 [2], explicitly cites arXiv:2603.24017 and restates the target as Conjecture 1 on PDF pages 1–2. Theorem 2 on PDF page 2 says:

> Conjecture 1 holds for all $d \geq 4$.

More explicitly, Zhang proves

$$\min_{0\neq x\in L} \frac{\|x\|_p}{\|x\|_2} = 2^{1/p-1/2} \qquad (0 < p \leq 1),$$

$$\min_{0\neq x\in L} \frac{\|x\|_p}{\|x\|_2} = \min\left\{2^{1/p-1/2}, \left(\frac{(d-1)^{p/2}+(d-1)^{1-p/2}}{d^{p/2}}\right)^{1/p}\right\} \qquad (1 < p < 2),$$

$$\max_{0\neq x\in L} \frac{\|x\|_q}{\|x\|_2} = \max\left\{2^{1/q-1/2}, \left(\frac{(d-1)^{q/2}+(d-1)^{1-q/2}}{d^{q/2}}\right)^{1/q}\right\} \qquad (2 < q < \infty).$$

Taking $p = 2\alpha$ for $\alpha < 1$ and $q = 2\alpha$ for $\alpha > 1$ gives exactly the source formulas after raising to the power 2α. The point $\alpha = 1$ is tautological. Thus Zhang supplies all $d \geq 4$, and the source's $d = 3$ theorem supplies the only remaining nontrivial dimension.

Provenance and scope

This is an explicit answer, not an implication discovered only after an unrelated theorem: the supporting paper names the Holevo–Utkin conjecture in its title, cites the source throughout the statement, and calls Theorem 2 the proof of its remaining cases. A direct exact-title and exact-formula search located arXiv:2605.05243. No part of the source conjecture remains open.

References

[1] A. S. Holevo and A. V. Utkin, *A conjecture on a tight norm inequality in the finite-dimensional* ℓ_p, arXiv:2603.24017.

[2] H. Zhang, *Proof of the Holevo–Utkin conjecture on sharp* ℓ_p *norms for zero-sum vectors*, arXiv:2605.05243v2.